



Statistical Process Control (SPC) Method to Improve the Capability Process of Drop Impact Resistance: A Case Study at Aluminum Cans Manufacturing Industry in Indonesia

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PAPER INFO	ABSTRACT
Chronicle: Received: 10 November 2019 Revised: 23 February 2020 Accepted: 25 March 2020	The purposes of this study are, first, analyzing why the Capability Process (Cpk) index of Drop Impact Resistance (DIR) does not meet the customer requirement or below 1.33, second, finding out what improvements should be made to make it meet the specifications. Statistical Process Control (SPC) is a method used in this research with supporting other tools such as Cause and Effect Diagrams (CED), Nominal Group Techniques (NGT) and why, what, where, when, and how (5W1H) method. After improvement was made, the results of the study were satisfactory, the average of DIR test was increased by 26.21% i.e. from 20.41 cm to 25.76 cm, standard deviations reduced from 1.80 to 1.48, and the potential Cpk increased significantly from 0.48 to 1.79. The SPC supported by other statistical tools was effective and efficient to improve quality, so the process statistically can be categorized as in control. At the end of this research, the further discussion is still needed to maintain what has been achieved and also to make the research even better. It is recommended for further researchers to examine the chemical composition of aluminum M₁ alloy 3104 influence on the strength of the drop impact resistance and to analyze data. It is recommended to integrate or combine SPC with Six Sigma or Engineering Process Control (EPC).
Keywords: Statistical Process Control (SPC). Capability Improvement. X-R Chart. CED and NGT. Cpk.	

1. Introduction

Recently, many the packaging industry were growing up in the world with the great opportunities, one of them is the aluminum cans packaging industry. The global demand for aluminum packaging in 2019 was predicted 335 billion cans as reported by Can Maker Magazine. In Indonesia, as stated by the Minister of Industry on www.Tempo.co.id edition 20 April, 2019, that investment in the food and beverage industry sector was increased by 11% in year 2019 compared to the previous year. In order to be able to capture those good market opportunities, the quality and price are the essential thing needs to be prepared. For getting it, the effort made by one of the aluminum cans packaging industry in Indonesia was doing innovation the aluminum raw material from type M₀ to M₁ with minor changing on the process to get the finished good cans quality both for visual and strength performance that are still on the specification. The constraint was noticed that the Cpk index for DIR does not meet the customer requirement or below 1.33. Fig. 1 is showing up of DIR test schematic for aluminum beverages

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cans, the test basically to give assurance if the aluminum cans after filling are safe during the transportation process.

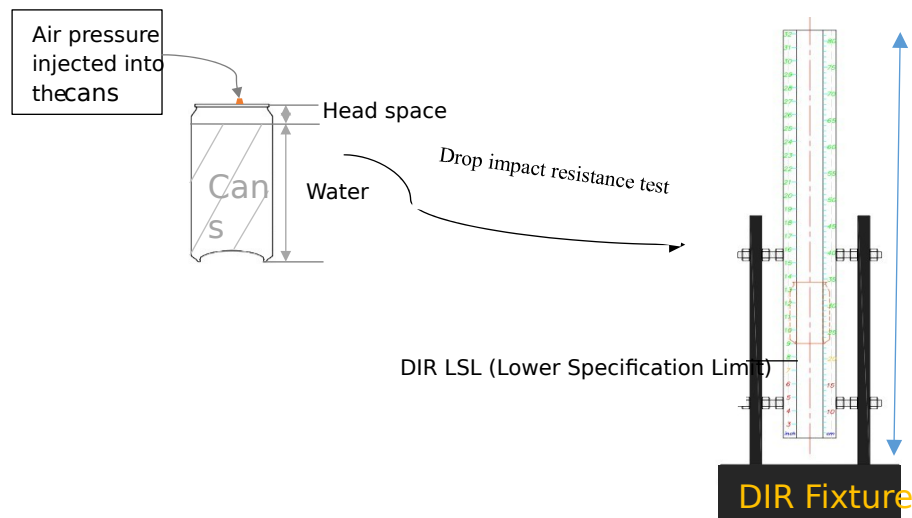


Fig 1 DIR test schematic.

The test result of nine quality parameters are as in Table 2 below, it is describing the achievement of the Cpk with M_1 .

Table 1. The 5 hours capability study data of aluminum cans Slk 330 mL with material M_1 .

Quality Parameters (unit)	Sample (n)	Ave.	Min	Max.	Std dev.	Cp	Cpk
1. Finish can height (mm)	180	146.02	145.87	146.15	0.05	3.04	2.24
2. Flange width (mm)	180	2.09	1.99	2.19	0.04	2.06	1.96
3. Plug diameter (mm)	60	50.05	50.01	50.10	0.002	3.40	2.50
4. Axial load (Lbs)	150	227.8	224	232	1.37	NA	12.60
5. Buckle strength (Psi)	150	96.12	94.4	98.3	0.86	NA	2.35
6. Thin wall thickness (mm)	200	0.092	0.09	0.095	0.001	NA	2.01
7. Dome depth (mm)	300	10.41	10.35	10.46	0.02	3.94	1.96
8. Reform diameter (mm)	300	44.78	44.76	44.82	0.01	3.92	3.24
9. DIR (mm)	150	20.41	17.28	22.86	1.80	NA	0.48

This study emphasizes the area of future research SPC with other tools to analyze and streamline process capability. Hence, this research is considered to advance and embrace high valued methodologies to help the industry to become more competitive in the global.

The aim of this study first is to analyze what are the causative factors which were causing the index Cpk could not meet the requirement or below 1.33; second, to knowing what the improvement should be made on the production process to make it meet the requirement by utilizing SPC with supporting other statistical tools.

2. Literature Review

SPC is a good method to measure and control the product quality during the process. As well as, aluminum cans packaging industry which produced beverages and beer cans needs to be controlled well in terms of the quality to give assurance that the product filled inside the cans is safe. The visual

aspect of the body of the cans is very important for the quality and it may conduct a serious defect that can affect the innocuity of the filled beverage [1]. If the packaging quality like aluminum cans categorized as good quality if it can be able to retain the carbon dioxide dissolved into the product for certain periods [2]. The packaging with good in form and design combining with proper material selection will be able to attract the customer to buying [3]. Page et al. [4] mentioned that the packaging functions must continue to be performed satisfactorily until well after the end of the stated shelf-life period.

Making sure the product quality is good or meeting the customer requirement its needs to be controlled well in every step of the production operation. SPC method can be implemented to maintain the process variations. Serious implementation of SPC to monitor process stability and capability through the identification of the root cause will create a positive result for the company [5]. As succeed implemented at molding plastic industry, SPC gave a significant improvement on the Cpk with increasing from 0.63 to 1.59 [6]. By supporting others' methods in determining the cause of process variation, SPC becoming more useful in the operation. As mentioned by Harpreet et al. [7] that SPC beside to control variation also proven as a technique for determining a process is called "under control" when process capability and predicting are yielded from a process.

To evaluate and analyze the production process, SPC can be used [8, 9]. SPC method improve the quality in mass production, and the process capability index is tool for continuous improvement [10]. Implementing SPC method with using fish born and process monitoring chart Cpk can be improved [11]. Cp and Cpk value beside to control process variation also can be used to monitor the quality control inter company or organization like studies conducted by Chen et al. [12]. To determining the root cause of the faulty in the process, Pareto chart, fish bone diagram and 5 whys can be utilized [13]. SPC have the power in ability to examine a process and sources variation in the process with only beneficiary two of seven quality tools were used as succeed implemented in automotive industry [14]. SPC method with applied $\bar{X}$ -R chart in monitor process stability that the significant result was achieved as study done by Ucurum et al. [15] at Cast Iron Part Production. Brainstorming with technical member in manufacture is effective for gather the idea for improvement the NGT method can be used as tools [16]. Brainstorming collected in NGT method; idea improvement based on the customer demand of Kaizen concept is be able to make it happen [17]. For reducing the losses on the process of Plan-DoCheck and Action (PDCA) with combining others method can be applied [18]. Kaizen concept with 8 steps of PDCA was success gratefully to reduce defect in automotive battery manufacturing [19]. To maintain the Cpk keeping on the track $\bar{X}$ -R chart can be applied [20].

3. Research Methodology

Potential capability index improvement process needs to go through several steps. Our study was organized over the beverage cans manufacturing industry. *Table 1* is representing as an initial capability study data achieved after changing input aluminum material to M₁.

Steps in overall Cpk improvement are as follows:

Step 1: Data collection.

Step 2: Data analysis [$\bar{X}$ -R chart & potential Cpk].

Step 3: Determine main root cause and dominant root cause (CED and NGT).

Step 4: Determine improvement with 5W1H.

Step 5: Standardization.

For detail, the steps of improvement process to improve potential Cpk are shown in *Fig. 4*.

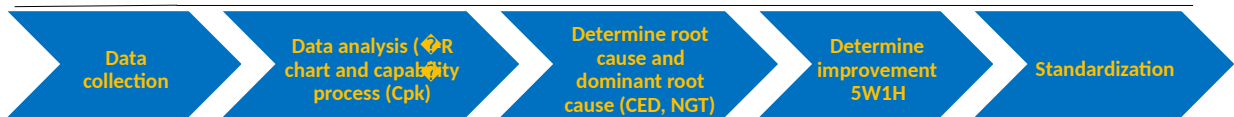


Fig. 2. Process capability improvement steps.

4. Findings and Data Analysis

As described in Table 1, the drop impact resistance is as one of critical parameter; after changing input material to material M_1 , the capability index does not meet the customer requirement as plotted to the run chart of the capability index achievement (Fig. 3).

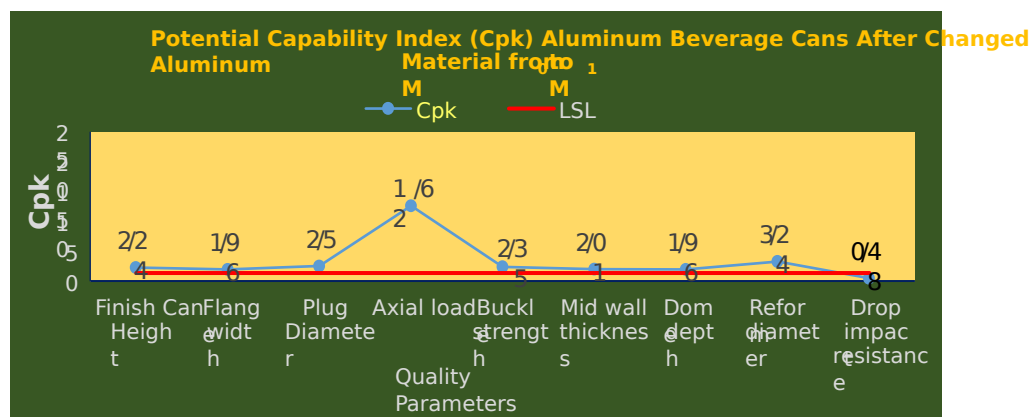


Fig. 3. Run chart of capability index achievement with material M_1 .

4.1. Data Analysis (X-R Chart and Cpk)

The quality parameter of DIR with index Cpk achievement was 0.48. Details of the control chart and histogram for DIR are shown in Fig. 4 and Fig. 5 in below.

With refer to Fig. 4 and Fig. 5, DIR with material M_1 both for stability and Cpk are does not meet the customer requirement. In term of stability process, on the $\bar{X}$ -chart, there were 4 points out of the control limits and on R-chart, there was one point touched the lower control limit, and the potential Cpk was 0.48 with mean the process was not capable. With these achievements, the research is needed; the aim of the research is to analyze the factors are causing the faulty of achievement for the Cpk index and to determining the steps of correction to improve the Cpk index to a minimum 1.33. To achieve all these objectives, each stage of the process needs to be controlled well; to control the quality at each stage of the process of aluminum cans forming, the method can be used is SPC, with supporting other methods and tools like CED, NGT, and 5W1H that will result positive improvement in term of the stability and capability of the process.

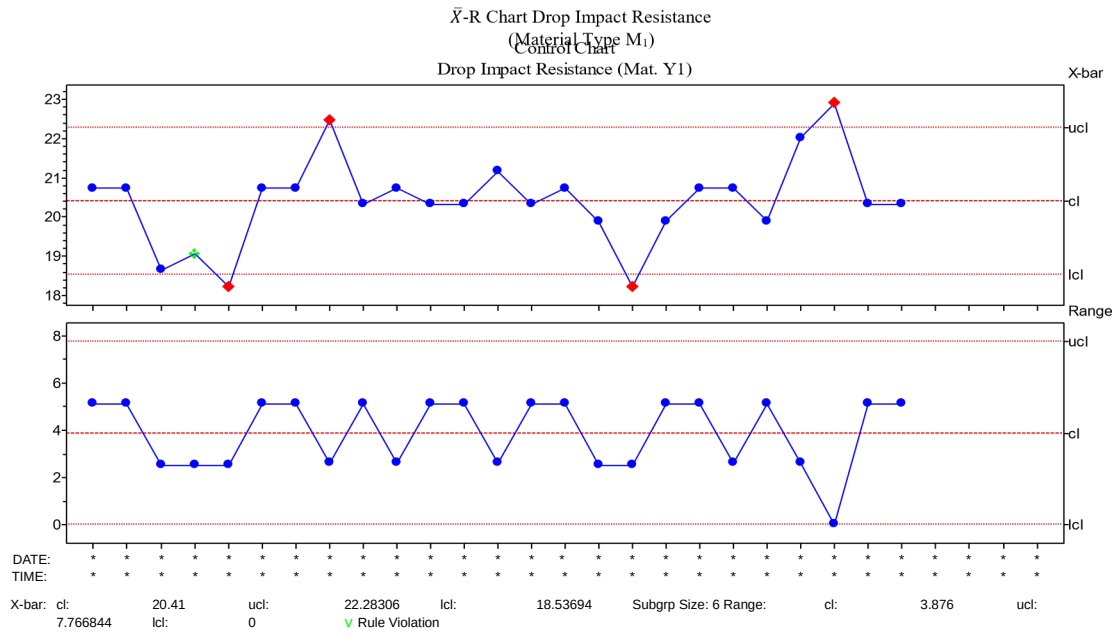


Fig. 4. Control chart drop impact resistance aluminum cans with material M_1 .

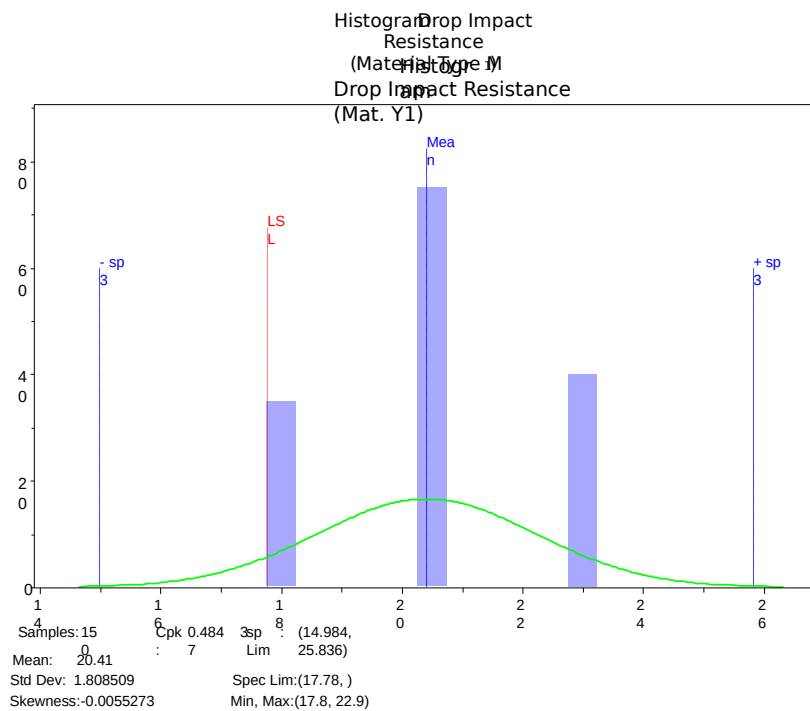


Fig. 5. Histogram of DIR for aluminum cans with material M_1 .

4.2. Determine route Cause and Dominant Root Cause (CED and NGT)

Before making CED, brainstorming has conducted with production, QA, Engineering, and Corporate production team to determine the main root cause, the brainstorming result as shown in Table 2.

Table 2. Brainstorming data to find the possible root cause of drop impact resistance faulty in the achievement of the Cpk index.

No	Causes	Causes Factor	Code
1	Annealing or softening (yield strength) of material aluminum.	Material	F1
2	Aluminum material thickness.	Material	F1
3	Washer oven dryer temperature.	Machine	F2
4	Temperature feco oven decorator.	Machine	F2
5	Temperature oven Inner Bake Oven (IBO).	Machine	F2
6	Mat conveyor jam with full cans inside oven dryer (Washer, Feco Deco or IBO) with duration > 5 minutes.	Machine	F2
7	The dimension of dome depth.	Machine	F2
8	The aluminum thickness dome area.	Tooling	F3
9	Profile/Geometry tooling: punch sleeve & dome die.	Tooling	F3
10	Dimension reform diameter.	Machine	F2
11	Air pressure that injected to inside the cans during testing DIR.	Method	F4
12	Base plate thickness for testing drop impact resistance.	Method	F4
13	Operator less knowledge.	Man	F5
14	Mistake or wrong in measurement.	Man	F5
15	Lack of lighting.	Environment	F6
16	Unstable speed of body maker machine.	Machine	F2
17	SOP not updated.	Method	F4

From 17 items of possibilities were causing for DIR do not meet the customer specification in term of the Cpk achievement, for making clear and easier in analysis then next table will give the classification information in more specific and details as stratification.

Table 3. The cause of the human (man) factor.

No.	Potential Cause	Causes Factor
13	Operator less knowledge.	Man
14	Wrong in measurement.	Man

The cause of the material factor is in Table 4.

Table 4. Cause of material factor.

No.	Potential Cause	Cause Factor
1	Annealing or softening material aluminum deformation after washer dryer.	Material
2	Thickness off aluminum material.	Material

The cause of the method factor is disprove in Table 5.

Table 5. Cause of method factor.

No	Potential Cause	Cause Factor
11	Air pressure that injected to inside the cans fluctuated.	Method
12	Base plate fixture of drop impact resistance.	Method
17	SOP not update.	Method

The cause of the machine factor is shown in Table 6.

Table 6. Cause of machine factor.

No.	Potential Cause	Cause Factor
1	Body maker machine speed fluctuated.	Machine
2	Temperature dryer washer.	Machine
3	Temperature feco deco.	Machine
4	Temperature IBO.	Machine
5	Mat oven dyer washer, feco or IBO jam or stopped > 5 minutes.	Machine

6	The dimension of dome depth.	Machine
7	The dimension of reform diameter.	Machine

The cause of the tooling geometry factor is shown in Table 7.

Table 7. Cause of tooling factor.

No.	Potential Factor	Cause Factor
1	Punch sleeve tooling geometry.	Tooling
2	The aluminum thickness dome area.	Tooling

The cause of the environment factor is shown in Table 8.

Table 8. Cause of environment factor.

No.	Potential Factor	Cause Factor
1	Lack of lighting	Environment

4.2.1. CED

CED or fish born diagram for DIR is shown in Fig. 6.

There were ten possible root causes of the Cpk why do not meet the customer requirement as can be seen on the red rectangular box with dashed lines as shown in Fig. 6. Details description is as follows:

- ☐ Cause factor of man: The possibility of operator did wrong or a mistake in measurement and lack of knowledge.
- ☐ Cause factor of material: YS of aluminum after washer dryer, the variation between before and after drying process was too high and the thickness of aluminum raw material.
- ☐ Cause factor of method: Air pressure that injected inside the cans > 60 psi and Base plate fixture for DIR test the thickness > 31 mm.
- ☐ Cause factor of the machine: Temperature oven washer dryer; Mat conveyor washer dryer, Feco oven deco or IBO jam or stopped > 5 minutes.
- ☐ Cause factor of tooling: Punch sleeve tooling geometry and aluminum thickness at dome area.
- ☐ Environment: Lack of light sources.

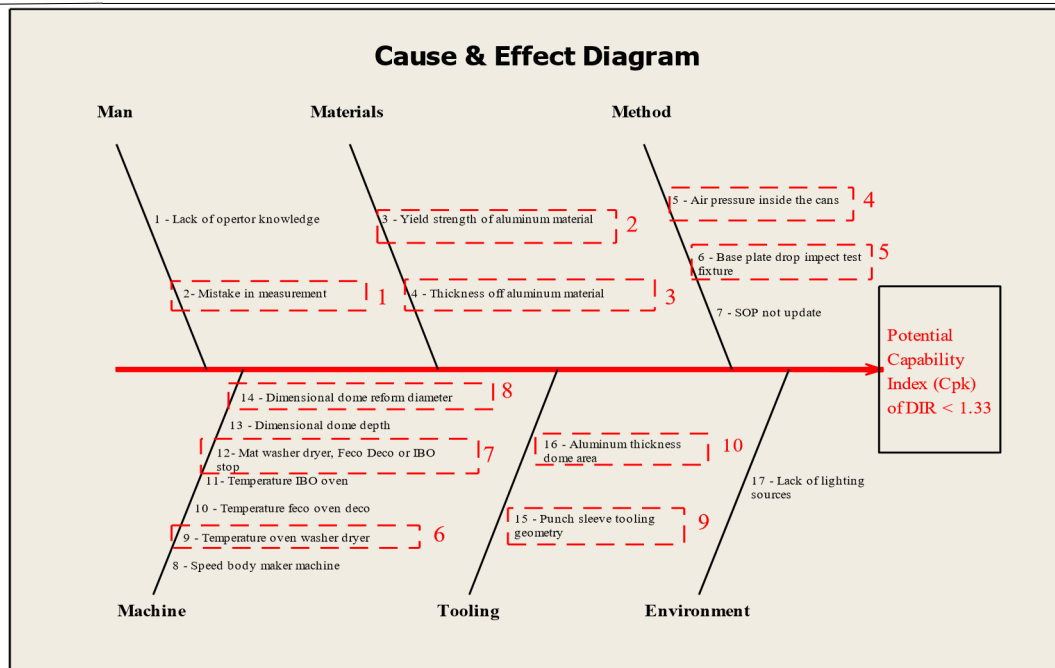


Fig. 6. CED or fishbone diagram for DIR faulty in achieving Cpk index.

4.2.2. Constructing NGT

From six factors with ten findings, then continuing to find out what the dominant cause is. NGT method is used, elaborating with 8 members of an expert at the plant with a different background as such education, age, year of service, and current expertise with these differences in the various background it will be resulted more accurate to give the information and finally correct decision is gotten. The concept of it is shown in Table 9.

Table 9. Data NGT total score from 8 members.

No.	Variable Causes	Scorer	Total Score
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		Scorer 1	Scorer 2	Scorer 3	Scorer 4	Scorer 5	Scorer 6	Scorer 7	Scorer 8	
1	VC ₁	4	5	5	4	5	6	5	5	39
2	VC ₂	5	4	6	5	5	5	5	5	40
3	VC ₃	5	4	5	4	5	5	6	5	39
4	VC ₄	7	8	7	6	5	8	7	8	56
5	VC ₅	5	5	4	6	4	5	5	6	40
6	VC ₆	7	5	5	6	6	7	7	8	51
7	VC ₇	5	5	5	5	4	5	6	5	40
8	VC ₈	5	4	5	4	5	6	5	5	39
9	VC ₉	8	7	8	6	7	8	7	7	58
10	VC ₁₀	5	5	4	5	5	5	6	5	40

VC₁: operator wrong measurement, VC₂: aluminum yield strength, VC₃: Aluminum material thickness, VC₄: air pressure that injected inside the cans for test DIR high fluctuations, VC₅: baseplate fixture drop impact resistance > 31 mm, VC₆: temperature oven washer dryer too high > 420°F, VC₇: Mat (oven dryer, IBO) and feco deco stopped for more than 5 minutes, VC₈: reform diameter dimensional, VC₉: tooling geometry of punch sleeve, and VC₁₀: aluminum thickness dome area.

NGT calculation is based on the basic comment calculation with the following formula:

$$NGT \geq \frac{1}{2}(\text{Total number of scorer} * \text{Caused Variable}) + 1. \quad (1)$$

On this study, number of the scorers are 8 and variable causes are 10, so based on the formula Eq. (1) the NGT will be as follows:

$$NGT \geq \frac{1}{2} (8 * 10) + 1, \text{ so } NGT \geq 41.$$

Revert to Table 9 linked to the NGT value, three potential variables having NGT value higher than 41, it becomes a dominant factor of the cause for the Cpk of DIR achievement, namely air pressure injected to inside the cans, temperature of oven dryer, and tooling geometry of punch sleeve.

5. Determining Improvement

5.1. Improvement Plan

Improvement plant for increasing the potential capability of DIR with 5W1H is shown in Table 10.

Table 10. Sequence of Cpk improvement for DIR (Plan).

No	Cause	Why	What	Where	When	Who	How
1	Unstable air pressure which was injected into inside the cans some time more than 60 Psi was observed.	Pressure gauge indicator was broken.	The pointer scale has not precise.	Regulator drop test fixture.	Aug 2019	Hadi.P	Changed with a new one.
2	The temperature of washer oven dryer too high > 420°F.	Only using one zone for drying the cans.	Reducing the temperature to below 420°F.	Oven washer dryer.	July 2019	Farid	Activated oven zone 2 to getting temperature oven dryer below 420°F.
3	Profile or geometry tooling of Punch Sleeve.	The clearance needs to be adjusted in matching with new material.	Punch sleeve nose radius.	Punch Nose Radius R_1 dan R_2	Sept 2019	Anton	Modifying punch nose radius, R_1 punch nose radius was changed from 0.05 inch to 0.06 inch and R_2 punch nose was changed from 0.042 inches to 0.05 inch.

After improvement plan as shown in Table 10, then we continue to implement improvement. 5.2.

Implementation of Improvement (Do)

Table 11. Implementation of quality improvement.

No	Implementation of Improvement	Improvement Activity
1	Air pressure is injected inside the cans before the test DIR.	To avoid air pressure that injected to inside the cans before test DIR, the pressure gauge indicator changed to the new one and doing a routine check for the function of pressure gauge with monthly bases and put on the calibration schedule.
2	Temperature oven washer dryer too high > 420°F.	When run production with material M_1 changed drying cans process from one zone to two zones with activated oven dryer zone 2, with set point temperature zone 1: 385°F and zone 2: 395°F.
3	Tooling geometry or profile punch sleeve.	Stretching aluminum at dome area point 3 (p3) as shown in Fig. 8 was reached 3.83% from the original thickness, it resulted in the drop impact resistance became weak.

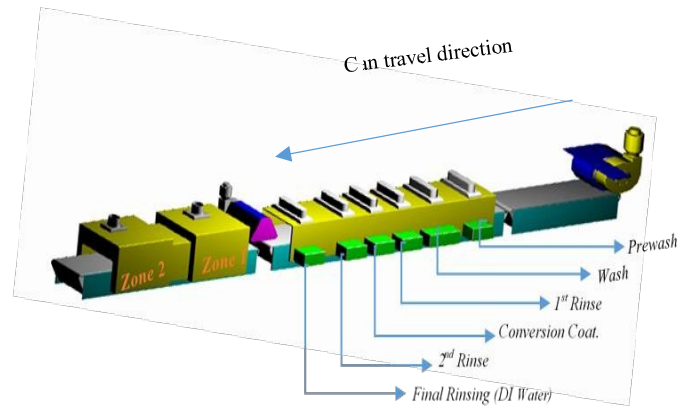


Fig.7 Oven washer dryer zone 1 and zone 2.

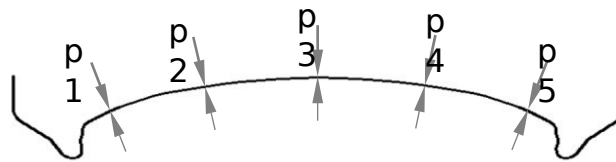


Fig. 8. Measurement point for aluminum thickness dome area of aluminum beverage cans.

To avoid stretching on the dome area p3, punch sleeve nose radius was modified, the modification area is shown in Fig. 9 in below.

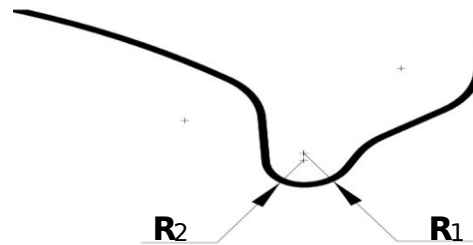


Fig. 9. Schematic of dome formation for aluminum beverages cans: R_1 : radius 1 and R_2 : radius 2.

Details for radius punch sleeve changing are as follows:

- Punch sleeve nose radius R_1 changed from 0.05 inch to 0,060 inch.
- Punch sleeve nose radius R_2 changed from 0.42 inch to 0,50 inch.

6. Evaluation (Check)

After some improvement were done then we follow the evaluation or controlling of improvement whether effective or not. The tools are used to do evaluation are $\bar{X}$ -R chart and Cpk.

6.1. Constructing $\bar{X}$ -R Chart of DIR after DID Some Improvements

Table 12 is DIR test data after some improvement done.

Table 12. DIR test data.

Date	Time	Machine	i	ii	iii	iv	v	vi	X	R
03 Oct, 2019	8:15	1	27.94	25.4	25.4	25.4	27.94	25.4	26.25	2.54
		2	25.4	27.94	25.4	25.4	27.94	25.4	26.25	2.54
		3	27.94	25.4	22.86	25.4	25.4	25.4	25.4	5.08
		4	25.4	25.4	22.86	27.94	25.4	27.94	25.82	5.08
		5	25.4	25.4	25.4	25.4	27.94	27.94	26.25	2.54
	9:15	1	27.4	25.4	27.94	25.4	25.4	25.4	26.16	2.54
		2	25.4	25.4	22.86	25.4	27.94	25.4	25.4	5.08
		3	22.86	25.4	25.4	27.94	25.4	25.4	25.4	5.08
		4	27.94	25.4	25.4	22.86	27.94	25.4	25.82	5.08
		5	25.4	25.4	27.94	25.4	25.4	25.4	25.82	2.54
	10:15	1	25.4	27.4	25.4	27.94	25.4	25.4	26.16	2.54
		2	25.4	27.94	25.4	22.86	25.4	25.4	25.4	5.08
		3	25.4	25.4	25.4	25.4	27.94	25.4	25.82	2.54
		4	25.4	25.4	27.94	25.4	27.94	25.4	26.25	2.54
		5	22.86	25.4	25.4	25.4	27.94	25.4	25.4	5.08
	11:15	1	25.4	25.4	27.94	25.4	25.4	25.4	25.82	2.54
		2	25.4	27.94	22.86	25.4	25.4	25.4	25.4	5.08
		3	25.4	25.4	27.94	25.4	25.4	25.4	25.82	2.54
		4	25.4	25.4	25.4	27.94	22.86	25.4	25.4	5.08
		5	25.4	25.4	22.86	25.4	27.94	25.4	25.4	5.08
	12:15	1	27.94	25.4	25.4	25.4	25.4	25.4	25.82	2.54
		2	22.86	25.4	25.4	25.4	25.4	27.94	25.4	5.08
		3	25.4	25.4	25.4	25.4	27.94	25.4	25.82	2.54
		4	25.4	22.86	25.4	25.4	25.4	27.94	25.4	5.08
		5	27.94	25.4	25.4	25.4	27.94	25.4	26.25	2.54
$\bar{X}$										
R										
25.77 3.76										

Based on the customer information, stated to the manufacturing quality plan that DIR specification was 17.78 cm minimum average.

6.1.1. Determining Center Line (CL), upper control Limit (UCL), and lower control Limit (LCL) of $\bar{X}$ chart

With refer to Table 12, following are how to determine CL, UCL, and LCL of $\bar{X}$ -Chart.

$$CL = \bar{\bar{X}} = 25.765. \quad (2)$$

$$UCL = \bar{\bar{X}} + A_2\bar{R} = 25.765 + 0.483(3.759) = 27.580. \quad (3)$$

$$LCL = \bar{\bar{X}} - A_2\bar{R} = 25.765 - 0.483(3.759) = 23.949. \quad (4)$$

A_2 is constant as shown in Table 13.

6.1.2. Determining CL, UCL, and LCL of R-chart

$$CL = \bar{\bar{R}} = 3.759. \quad (5)$$

$$UCL = D_4\bar{R} = 2.004(3.759) = 7.533. \quad (6)$$

$$LCL = D_3\bar{R} = 0(3.759) = 0. \quad (7)$$

D_3 and D_4 are constants as shown in Table 13.

Table 13. The control chart constants.

Sample Size = m	A2	A3	d2	D3	D4
2	1.880	2.659	1.128	0	3.267
3	1.023	1.952	1.693	0	2.574
4	0.729	1.628	2.059	0	2.282
5	0.577	1.427	2.326	0	2.114
6	0.483	1.287	2.534	0	2.004
7	0.419	1.182	2.704	0	1.924

Calculated results from points 6.1.1 and 6.1.2 plotted to the chart are shown in Fig. 10.



The calculation results above plotted to the histogram by using NWA Quality Analyst v6.3 with the output is shown in *Fig. 11*.

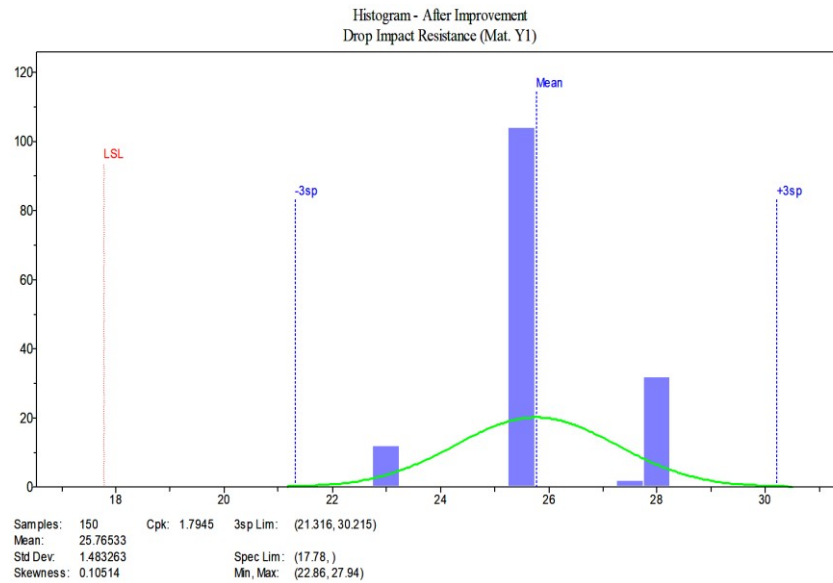


Fig. 11. Histogram of DIR with material M_1 after improvement.

Based on Fig. 11, the Cpk index has reached 1.73, it means the DIR has capable because of the index of $Cpk > 1.33$.

7. Standardizations (Action)

As described in the result and discussion above, the DIR after doing some improvement has already stable and capable if we plot on the distribution plot as shown in Fig. 12.

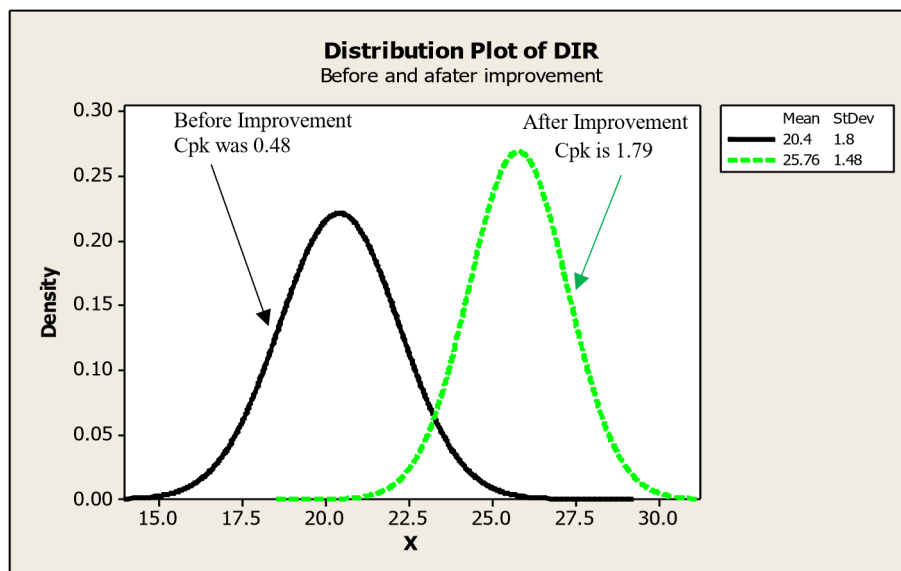


Fig. 12. Distribution plot of DIR before and after improvement.

With the above achievement, the next step is determining the standardization to maintain if the improvement has done, as guidelines in the future.

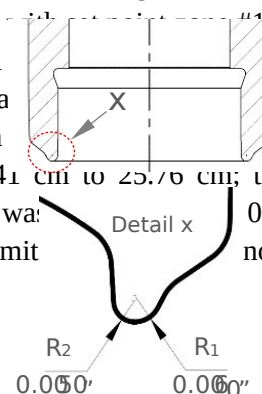
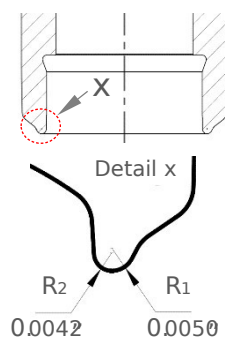
Table 14. Standardization to maintain the Cpk of DIR with material M_1 (Cont's).

Dominant Cause	Remarks
Air pressure was injected into the cans has fluctuated with the range 4 psi so the DIR test results becoming unstable	Before improvement Unstable reading on the pressure gauge indicator. After improvement Stable reading by changing the pressure gauge indicator and put on the permanent mounting. Standardizations Pressure gauge indicator, put in calibration schedule to make well control.
Temperature oven dryer washer setpoint over then 420°F.	Before improvement Cans drying process was using one oven zone with setpoint temperature 435°F with curing time 1 minute and 26 seconds.
The tooling geometry of the punch sleeve does not match for new aluminum type M ₁ .	Standardizations Issue oven card, monthly bases, and verified by engineering Dan quality assurance manager. Before improvement Punch Nose Radius R1: 0.050 Inch R2: 0.042 Inch
Cans drying process was using two oven zones with setpoint zone 1: 385°F and zone 2: 395°F, with curing time 2 minutes and 52 seconds.	Standardizations Revision Technical Drawing of Punch sleeve. Documents: 0106384, Rev 1 (29-07-19)

8. Conclusion

The root cause due three inside the cans range 4.0 psi; was 435oF, it's

softening when machine stops for a while; third, the punch sleeve nose radius tooling geometries, the clearance does not match with dome die. The improvement has been done with changing the pressure gauge indicator with putting on the rigid mounting and to avoid annealing or softening of the aluminum cans the oven dryer washer was activated with 385oF then the punch nose radius R1 correct clearance between punch nose and dome die; punch nose radius was changed from 0.050 inch to 0.060 inch and punch nose radius R2 was changed from 0.042 inch to 0.050 inch. After improvement was done the DIR increased by 26.21%, from 20.41 cm to 25.76 cm; the standard deviation was reduced from 1.80 to 1.48 and then the index Cpk was 0.48 to 1.79. Then, for the $\bar{X}$ -R chart and R-chart, point is out of control limit



statistically the process is categorized in control and capable. At the end of this research, further discussion is still needed to maintain what has been achieved and also to make the research even better, it is recommended for further researchers to examine the chemical composition of aluminum M1 alloy 3104 influence on the strength of the drop impact resistance and to analyze data it is recommended to integrate or combine SPC with Six Sigma or EPC.

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